

# Practical 5 - Non-proportional hazards and model assessment

## Survival Analysis

London School of Hygiene & Tropical Medicine

Datasets required: rotterdam (part of the survival package)

R packages required: survival, aod

```
library(survival)
library(aod)
```

## Introduction

In this practical we will use the methods introduced in the notes/lecture to assess some of the assumptions that underpin the Cox proportional hazards model. This practical uses the Rotterdam breast cancer study data, which contains information on 2982 individuals who were followed up after undergoing surgery for breast cancer. Several variables were collected on the participants at the start of follow-up. In this practical we will focus on the variables summarised in the table below. Our focus will be on the outcome of death. The models used in this practical are not designed to answer a specific question, but to help us practice using the techniques introduced in the lecture.

| Variable | Description                                              |
|----------|----------------------------------------------------------|
| pid      | Study ID number                                          |
| dtime    | Time to death or censoring in days                       |
| death    | Event indicator: 0=Censored, 1=Died                      |
| age      | Age at the start of follow-up, in years                  |
| hormon   | Indicator of receiving hormone therapy: 0=No, 1=Yes      |
| size     | Tumour size: 1: $\leq 20$ mm, 2: 20 – 50mm, 3: $> 50$ mm |

The data is part of the survival package so does not need to be loaded separately.

## Aims

The aims of this session are:

- Investigate the proportional hazards assumption using covariate-by-time interactions.
- Investigate the proportional hazards assumption using Schoenfeld residuals.
- Fit and interpret extended Cox models that relax the proportional hazards assumption.
- Use Martingale residuals to investigate suitable functional forms for continuous covariates.

## Question 1

- Load the data, and create a new time variable that measures time in years instead of days.
- Summarise the survival data: how many events are there and what is the range of follow-up times.
- Obtain a Kaplan-Meier plot for overall survival, and interpret.

## Question 2

- Fit a Cox proportional hazards model for the hazard of death, including the following explanatory variables: `age`, `hormon`, `size`.
- Interpret the hazard ratio estimates.

## Question 3

- Using the code below, reformat the data in such a way that each person's follow-up is divided into the following time periods (in years) up to their end of follow-up due to death or censoring: (0, 5], (5, 10], (10, 15], (15, 20]. We will refer to this as the 'split data'.

```
#Split the data into time periods
split.times1<-seq(5,20,5)
rdm.split1<-survSplit(Surv(dtime_yrs,death)~.,cut=split.times1,
                      data=rotterdam
                      ,start="start.time",end="stop.time",
                      event="death",episode="period")
```

- Take a look at the resulting data to understand how each person's follow-up is now formatted. How many rows are there in the split data?

- Refit the Cox model fitted in question 2 using the split data. To do this we need to take into account that each row in the data now represents a different period of follow-up, with a different start time and end time. This needs to be accounted for when we fit the model, through how we specify `Surv(time,time2,event)`.
- Check that you get the same results from this model as you got in question 2.

## Question 4

- Return to the original data. Reformat the data again in such a way that each person's follow-up is divided into small periods that are divided by the observed event times. Modify the code above to do this.
- Take a look at the resulting data to understand how each person's follow-up is now formatted. How many rows are there now in the split data?
- Refit the Cox model fitted in question 2 using the split data, using the same approach as used in question 3.
- Check that you get the same results from this model as you got in questions 2 and 3.

## Question 5

Using the split data created in question 4, we will assess the proportional hazards assumption for the `hormon` variable by adding an interaction between age and time in the Cox model. The 'time' variable that you use here should be `stop.time`, which is the end time of each small interval.

- Write down the form for the hazard model being fitted.
- Fit the model and interpret the results.
- Repeat for each variable included in the model, i.e. including an interaction between each variable and time.
- Assess the null hypothesis of proportionality of hazards for each variable.

## Question 6

Return to the original data. There is a ready-made function that allows you to fit the models in question 5, with covariate-by-time interactions, without having to split the data on all event times first (the function does this for you in the background). Try out the code below and see if you get the same results as when you did this manually in question 5.

```
rdmtt.age.2<-coxph(Surv(dtime_yrs,death)~age+hormon+as.factor(size)+tt(age),
                    data=rotterdam,
                    tt=function(x,t,...){x*t})
summary(rdmtt.age.2)
```

## Question 7

Use the original data (i.e. not split on time)

In this question we will look at the Schoenfeld residuals for assessing the proportional hazards assumption.

- Based on the Cox model you already fitted in question 2, plot the Schoenfeld residuals for each variable using the code below. Assess the results.
- Compare your findings with those from question 5 (and 6).

```
#Cox model
rotterdam$size_2050<-ifelse(rotterdam$size=="20-50",1,0)
rotterdam$size_50plus<-ifelse(rotterdam$size==">50",1,0)
rdm.cox<-coxph(Surv(dtime_yrs,death)~age+hormon+size_2050+size_50plus,
               data=rotterdam)

#Schoenfeld residuals
sch.resid<-cox.zph(rdm.cox, transform = 'identity')
sch.resid

par(mfrow=c(2,2))
plot(sch.resid,col="red",lwd=2)
```

## Question 8

You may have found some evidence that the proportional hazards assumption may not be appropriate for **age** and **size**. In this question we will use an extended Cox model to relax the proportional hazards assumption for **age**.

- Go back to the split data that you created in question 3, where each person's follow-up was divided into 5 year periods. Fit an extended Cox model, allowing for an interaction between time period and age.
- What are the estimated hazard ratios for age in each of the four time periods?

## Question 9

Use the original data (i.e. not split on time)

In this question we will relax the proportional hazards assumption for **size** by using a stratified Cox model.

- Write down the form of the Cox hazard model, assuming proportional hazards for **age** and **hormon** and a separate baseline hazard for each category of the size variable.
- Fit the model using the `strata()` option in `coxph`.
- Interpret the results.

## Question 10

### Use the original (i.e. not split on time) data

In this question we will use Martingale residuals to investigate whether the functional form for our continuous variable **age** is appropriate, or whether transforming age might provide a better fit.

- Fit a Cox proportional hazards model including **hormon** and **size**, but *excluding* **age**. Use the code below to obtain the Martingale residuals from the model and plot them against age. What is the association between the residuals and age and what does this suggest to you about what an appropriate functional form might be for age in the model?

```
#Martingale residual plot for age: excluding age from Cox model
rdm.cox.noage<-coxph(Surv(time=dtime_yrs,event=death)~
                    hormon+as.factor(size),data=rotterdam)
mgale.res.noage<-resid(rdm.cox.noage,type="martingale")

plot(rotterdam$age,mgale.res.noage,
     xlab="Age",ylab="Martingale residuals")
lines(lowess(rotterdam$age,mgale.res.noage),col="red",lwd=2)
abline(h=0,lwd=2,col="grey")
```

- Fit a Cox proportional hazards model including **age**, **hormon** and **size**. Obtain the Martingale residuals from the model and plot them against age. What is the association between the residuals and age and what does this suggest to you about what an appropriate functional form might be for age in the model?
- If you think a transformation of age might improve the model fit, do this and reassess the model fit.